ADUPA NITHIN SAI

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EDUCATION

Master of Science in Artificial Intelligence, *University of Michigan-Dearborn* **2024 - 2026**

Bachelor of Technology in Artificial Intelligence & Machine Learning, *JNTUH* **2020 - 2024**

EXPERIENCE

AI Engineer, *SteadfastAI* **May 2025 - Present**

- Designed and launched high-accuracy (90%+) cybersecurity ML models, fully integrated into the MCP (Microservices Control Platform) for real-time, automated threat detection and incident response.
- Engineered scalable, cloud-based AI APIs and microservices, cutting deployment time by 30% while enhancing system performance, reliability, and cross-platform integration.

Research Assistant, *University of Michigan-Dearborn* **Nov 2024 - Present**

- Leading the AI Transformation initiative with researchers, increasing scientific research productivity by 20% and optimizing AI models using Python by 15%. Leading the FakeXpose team, where this research directly enhances the web app by analyzing results, improving models, and refining user experience.

AI Developer, *Plausibility Solutions* **Nov 2023 - May 2024**

- Engineered and delivered machine learning models that improved predictive accuracy by 30%, directly contributing to more accurate business decisions by automating the hiring process.
- Created and implemented AI algorithms to automate business processes, reducing operational time by 25%.
- Led the development of an AI-based system that streamlined client operations, improving efficiency by 20%.

Devops Engineer Intern, *MoxieHawk Pvt Ltd* **Apr 2023 - Oct 2023**

- Configured and optimized CI/CD pipelines using Jenkins, reducing deployment time by 40% and enabling faster releases.
- Spearheaded the deployment of scalable AI architecture, facilitating the integration of 5 new AI-expert models.

Developer and Test Administrator Intern, *Eunoia Innovations Pvt Ltd* **Mar 2022 - Feb 2023**

- Coordinated the deployment of intelligent system boats for automated data collection, improving operational efficiency by 40%. Conducted testing & optimization of machine learning algorithms, increasing system reliability by 20%.
- Built a Remote-Operated Survey Boat for the Indian Maritime University (IMU), enhancing data collection capabilities in marine environments. Led cross-functional testing of Artificial Intelligence systems, improving the accuracy of data processing by 15% and ensuring reliable research outcomes.

SKILLS

Technical Proficiency: Python, JavaScript, Linux, Git, Flask, Django, Bootstrap, APIs, WordPress, Microsoft Office
AI & Machine Learning: Machine Learning, Deep Learning, Neural Networks, Computer Vision,PyTorch,TensorFlow, Keras
Cloud & DevOps: Vercel, AWS, Azure, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, MySQL, SQLite, RDBMS
API & Web: RESTful APIs, FastAPI, API Testing (Postman), Secure API Design
Data & Analysis: SQL, Pandas, Matplotlib, Plotly, advanced analytical techniques, data visualization
Security: Penetration Testing, Content Authenticity (C2PA), Data Encryption, Secure Systems Design

PROJECTS

- **Multi-Model Deepfake Detection System:** Assembled a high-accuracy deepfake detection system utilizing a multi-model approach to improve identification of AI-generated manipulations. Integrated custom-built and pre-trained models to boost detection accuracy by 30%, reduce false positives by 15%, and enhance media integrity against deepfake threats.
- **AI Classifier:** Designed and deployed a Python package using transformer models to classify text as AI-generated or human-written, accelerating classification speed by 25%.
- **Monitor-Load:** Developed and published a Python CLI tool to monitor CPU & GPU load and dynamically pause/resume processes, improving system resource management by 30%. Packaged for PyPI with pip install monitor-load.
- **NSAencrypt:** Launched a Python package for encrypting and decrypting multiple media formats. Attracted 50+ users.

PUBLICATIONS

Patents: Patent for an AI and Machine Learning algorithm to detect violence and non-violence percentages from real-time CCTV footage, achieving 90% accuracy (Patent Application No: 202341029626).

Research Papers:

- Data Encryption with RSA Cryptography, ICCCN 2024 - Explored secure frameworks for data sharing.
- Collecting, Curating, and Annotating Good Quality Speech Deepfake Dataset for Famous Figures: Interspeech 2025.

Journal Articles: A journal article on Machine Learning for the Detection of Violence from CCTV Live Footage in the Journal of Image Processing and Artificial Intelligence, achieving a 92% detection rate across 1,000 hours of footage.

Python Libraries: Created Deepfake Detector, a Python library to analyze and detect deep fake content in multimedia.